



Finance Video Q&A Bot

A Multimodal AI Chatbot for YouTube Video Q&A

João • Ironhack — AI Engineering Bootcamp • Final Project

Watch less. Know more.

Hours of personal-finance videos hold great advice — but who has time to watch them all? This chatbot lets you ask a question and get an answer grounded in real YouTube videos, with a link to the exact second it's discussed.

"Watch less. Know more."



Ask by text or voice

Speech-to-text turns your spoken question into a query



Answers from real videos

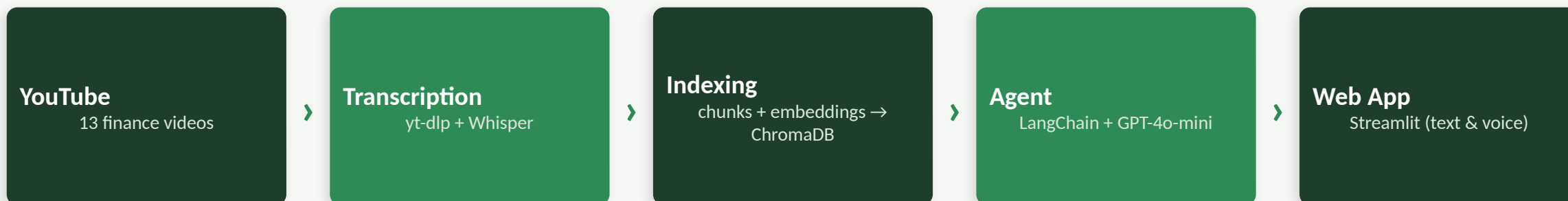
RAG retrieves the relevant moments — no hallucinations



Jump to the exact moment

Every answer links to the timestamp in the source video

How it all fits together



Offline (one-time): videos are transcribed and indexed into a vector database. **Online (live):** the agent retrieves the relevant chunks and GPT-4o-mini writes a grounded answer.

Deployment

- Deployed as a public web app on Streamlit Community Cloud, with secrets managed securely.
- Heavy models (Whisper) run offline for batch transcription; the live app uses lightweight OpenAI APIs — so the server stays small.

From video to searchable knowledge

Source & retrieval

- 13 personal-finance YouTube videos (investing, ETFs, budgeting, emergency funds, loans, dividends...)
- Audio downloaded with yt-dlp
- Transcribed with faster-whisper (small.en) — captions fallback (youtube-transcript-api)

Chunking

- Whisper segments grouped into ~1000-char chunks
- 150-char overlap so ideas aren't cut between chunks
- Each chunk keeps its start timestamp → deep links

Stored in the vector database (ChromaDB)

- Each chunk → an embedding (OpenAI text-embedding-3-small) for semantic search
- Metadata per chunk: video_id, title, channel, url, source, start_time
- ~190 chunks indexed across the 13 videos

An agent that reasons, not just retrieves

Architecture

- LangChain tool-calling agent (AgentExecutor)
- LLM: GPT-4o-mini (low cost, low latency)
- Decides which tool to use for each question

Memory

- Conversational memory via RunnableWithMessageHistory
- Per-session history → follow-up questions work
- e.g. "...and how much should I save?" keeps context

Prompt template (grounding & safety)

- Answer ONLY from retrieved context — say "I don't know" instead of guessing (anti-hallucination)
- Concise answers, always ending with: "This is educational content, not financial advice."
- Built with LangChain Expression Language (LCEL)

What the agent can use



search_finance_videos

Semantic search over the ChromaDB vector store (RAG)



list_available_videos

Lists the videos in the knowledge base



Speech-to-Text

OpenAI Whisper API transcribes the user's voice question



Text-to-Speech

OpenAI TTS (tts-1) reads the answer aloud

Measuring quality, not assuming it

1.00

Groundedness
(no hallucinations)

0.79

Correctness
(vs reference answers)

How

- LangSmith tracing + LLM-as-a-judge
- 14-question test set, 2 evaluators
- Latency & token cost tracked per run

Findings

- Perfect groundedness → the bot never invents answers; it declines when the videos don't cover a topic.
- Correctness gap is mostly coverage (declining on out-of-corpus topics), not wrong facts.
- Total evaluation cost: a fraction of a cent — GPT-4o-mini keeps it cheap.

Conclusion

A working, deployed multimodal RAG assistant

- ✓ End-to-end pipeline: YouTube → Whisper → ChromaDB → LangChain agent → web app
- ✓ Text & voice, with answers linked to the exact video moment
- ✓ Grounded and measured with LangSmith (no hallucinations)

Future work

Larger corpus · multi-language support · richer evaluation set · inline citations

LIVE DEMO

Let's try it.

 finance-ai-bot-finalproject.streamlit.app

A few sample questions ready to paste — backup screenshots prepared in case of network issues.